in Santa Catalina. If I had landed the plane, there's absolutely no way I would've forgotten that landing.

As soon as I thought that thought, I realized that my brain had been working very strangely. I realized that I'd been in a plane crash and it was real. And I just jerked my head up right away and realized that everything I was starting to suspect was real. My head started working immediately and retrieving and forming memories, I could feel it. And what was strange was, I could feel both states of mind. I had just come from a state where I wasn't forming memories, and now I was moving into this different state where I was forming memories. I could feel both states of mind at the same time, which was so strange.

Then I looked at the bed stand next to me, and there were something like a hundred cards from people I had received while I was in the hospital. They were sending me best wishes, saying get well and all that. And I read them. They were all from my very closest friends and associates.

And I said, Oh my god, I didn't even know they were there.

But I must have seen them every single night. Because they were there every single night. So it was like coming out of a very strange state and realizing that your head has not been forming any memories. That's what I deduced.

The very next day, my father called to remind me that I was supposed to show up for an appointment with the psychologist I'd been seeing. I had no memories of ever seeing a psychologist. But I went up to Stanford to see that psychologist and I kind of excitedly started explaining to him that I hadn't been forming memories or remembering the plane crash, and suddenly I'd come out of it. My head just switched over, I told him. It was amazing.

And would you believe it? He didn't believe me! I suppose I was so excited when I told him about this that he kept telling me I was a manic-depressive. I was stunned, and told him that I didn't have big highs or big lows like a manic-depressive would. I told him I was a very stable person. He said, "Well, manic depression usually starts when you're thirty." I was thirty. He had interpreted my excitement about my memory returning as being manic. What a quack.

Well, those five weeks after the plane crash, when I was finally and fully out of the amnesia, I decided this was a lucky opportunity. I should finish college, and not go back to Apple right away.

I realized it had been ten years since my third year of college, and if I didn't go back to finish up now, I probably never would. And it was that important to me. I wanted to finish. And I had already been out of Apple for a while anyway—five weeks without knowing it, actually—so that made it easier to just go back to school and not go back to Apple right away. I decided that life is short, right? So I decided.

I applied and got accepted and registered under the name Rocky Raccoon Clark. (Rocky Raccoon was the name of my dog, and Clark was my fiancée Candi's soon-to-be maiden name.)

And soon after I made that decision, Candi and I set the date to get married: June 13, 1981. It was an amazing party. We had the Apple hot-air balloon there in the front yard of Candi's parents' house. It was a spectacular party. Emmylou Harris, the famous folksinger, sang at the reception.

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The day after the wedding, I got an apartment in Berkeley to get ready to begin my fourth year of college. And on the weekend, the plan was that I would go back to this house we had bought on the summit of the Santa Cruz Mountains. It was amazing. Just a huge castle of a place.

It had a lot of flat land, which is unusual, so I had tennis courts built. And Candi turned a little pond into a nice little lake. I also bought an adjoining property, making twenty-six acres in all. It was a paradise. (Candi, now my ex-wife, still lives in that paradise.)

Candi stayed there working on the house while I spent the week in this college apartment a couple of hours north, in Berkeley. It was a great year, and a fun year. Because I was going under the name Rocky Raccoon Clark, no one knew who I was. I had fun posing as a nineteen-year-old college student, and the engineering classes were so easy for me. Every weekend, I went back home to the castle.

One of the first things I did at Berkeley, in addition to taking engineering courses for my degree, was to enroll in both psychology courses (for majors) and two courses specifically about human memory. After my accident and amnesia, I was intrigued by such strange aspects of memory, and I wanted to understand it more.

As far as my own condition went, it turned out to be relatively well known. It happens frequently to people after car and plane accidents, and it's associated with damage near the hippocampus section of the brain. It was a typical condition. There is no excuse for why my doctors—especially my psychologist—didn't figure this out.

Chapter 17

Have I Mentioned I Have the Voice of an Angel?

After the plane crash in 1981 and after I decided to go back and finish my degree at Berkeley, something else happened that I never would have expected.

It was during that first quarter at summer school when I was taking a class in statistics so I could enroll the following year. I was driving around in my car listening to a radio station—KFAT out of Gilroy, California—a station that had heavily influenced me during the Apple days. You see, I'd changed my music tastes from normal rock and roll to a type of really progressive country by then.

This was a new and strange type of music I'd never been exposed to before—a lot of folk, a lot of country, and a lot of comedy. It wasn't some dumb old countryish beat and song and themes; these songs were a lot about life. They very much reminded me of the sort of thinking Bob Dylan did, being as familiar with his lyrics as I was. And these songs went as deep—they pointed out what was right and wrong in life. The way they were written and the way I experienced them brought out a lot of emotion in me. I mean, there was a real meaning attached to these songs, and I was heavily influenced by this station.

At around this time, I recall seeing the movie *Woodstock*.

There was a meaning attached to that movie, too. A meaning that had to do with young people growing up and trying to find alternative ways of living. And so much of that was brought up in the words of these new progressive country songs I was listening to, like a music revolution was starting all over again.

And it hit me. I thought: Why not? Why not try to do a kind of Woodstock for my generation? I realized at this point that I had so much more money than I could ever dream of spending. I was thirty at the time and probably worth a hundred million dollars or more. I thought: My god, why not put on a big progressive country concert with these groups I loved? A lot of people might come.

At the time, I thought of it as kind of an unplanned event that would just happen.

Of course, I knew I didn't know enough to manage a concert or put one on. I didn't know the first thing about it. So I talked to a friend of mine, a friend who owned a nightclub in Santa Cruz called the Albatross, a strange name for a place like that. His name was Jim Valentine. I told him about my idea and convinced him that the kind of concert I had in mind would really draw a lot of people. Jim agreed, and man, it was nice to have one person agree with me. Most people didn't think progressive country could draw a crowd.

Now Jim, who owned that nightclub in Santa Cruz, also ran it. He had comedians on his stage, he had singers and songwriters come in, he had musicians play. And he had some connections to the early big music concerts—things like Altamont in 1969 and the early San Francisco Bill Graham days. So even though I had these connections, I thought, Well, maybe in a few years. I'll finish at Berkeley and then do it.

But then Jim called me and said he had a guy who could put this thing on. He said he'd found the one guy he knew who could organize and manage a project this large. But it was going to run many millions of dollars to create. That guy's name was Pete Ellis.

After talking about this to Jim, I realized this concert was going to be huge. Huge. We were envisioning a huge outdoor space where people could just drive up and camp out for three days, like a Woodstock thing. But maybe better.

By the time we'd gotten to this point, I'd already started going back to school. (And at school, remember, I'd tricked everyone into thinking I was a student named Rocky Raccoon Clark.) I'd also just gotten married to Candi, and we'd just bought that castle of a house—with the house number of 21435. (I liked that number mathematically because it had all the first five digits appear exactly once.)

Candi was also supportive of the idea of a concert, probably because her background was kind of a hippieish Grateful Dead thing. I told her I thought if enough people came, it would make money. I wasn't sure enough people would come, but I didn't care. I knew I could afford it. I didn't know how much money would come back, exactly, but I was willing to take the risk. And after I was introduced to Peter Ellis, he put out that it would take a budget of $2 million to get started, and I was willing to pay that.

For that money, the starting amount needed, I could basically form a corporation (the UNUSON Corporation, short for UNite Us in SONg), hire people, do the planning, get the site, and put the whole thing together.

I remember when he came up to my apartment in Berkeley on Euclid Avenue one evening. I presented him with a check for $2 million. Then he knew it was for real.

Well, I should mention here that two weeks after I wrote that $2 million check, I read a book called *Barefoot in Babylon*, by Bob Spitz, which was about the entire progression of creating Woodstock from day one. It was about finding staff, getting permission for sites, publicity, getting groups signed up, overcoming political hurdles, changing sites at the last minute, inadequate preparations for the numbers of people who would show up, and more mishaps. Every chapter took my breath away and had me thinking, Oh my god, what a disaster. That book really chilled me. I thought, What have I gotten myself into?

Let me tell you, if I had read that book two weeks earlier, I never would have done it. Period. I absolutely wouldn't have done it.

I mean, according to that book, Woodstock broke even only because of the movie. Also, the expenses involved in putting on Woodstock were small enough because they didn't do an adequate job of setting up for and handling a large audience. Had they spent that money, they would've lost everything. And Woodstock was a rainy, swampy mess. It wasn't what we all imagined after seeing the movie. In fact, in putting together the US Festival, I later did talk to one of the two guys who'd created Woodstock, and he didn't want to work with us. He'd consult, that's it. He didn't want to do it again. He said he was just a music company executive and it was kind of like they got started on this thing and ended up captives to it.

In a way, that happened to me. The US Festival was exactly the opposite of the Apple experience for me. It didn't come easily. It involved having plans to get certain groups, and having those groups cancel. It involved having plans for sites, and having those sites cancel. It involved having plans for equipment, and having the equipment not come through. It was a costly battle to do all the right things, but we did them anyway.

I'd written a check. I had confidence in my people. I'd already taken a stand, and when you take a stand, you don't back away from it. Sometimes this has been a big problem in my life—especially marriage-wise—but if I'm in, I'm in. I don't back out. And by the time I could see this was a disaster, I had this guy, Pete Ellis, and all the people he'd hired, counting on me. I couldn't just all of a sudden pull the rug out. And we'd already planned the date: the first US Festival would be the Labor Day weekend of 1982, right after my first year back at school.

We finally secured a site, a county park near San Bernardino. It was in kind of a depressed area. The county park needed money, and they saw us as a way to get those funds.

There were some great things about this site. For one thing, it was an enormous area which would let us bring lots of trucks and stuff into the amphitheater. This place had the capacity to easily hold about 400,000 people, and hopefully as many as a million. That's twenty times what the Shoreline Amphitheater in Mountain View holds. (I built Shoreline years later with concert promoter Bill Graham and heiress Ann Getty. I put in $3 million of the $7 million total.)

We didn't want to use a preexisting arena or stadium, we wanted more of a campout-style setup. And they had a lake and a big area. We had to groom it with all these trucks going day after day after day digging up dirt and getting the right shape. And then we had to quickly plant some fast-growing grass sod to create sort of a grass liner that would span many, many acres.

We of course had to plan for the huge number of people we thought would come. We actually even got a temporary freeway exit, and we got some top highway patrol people who were on our side. They got things approved. The sheriffs of San Bernardino County were behind us, too. We were given this kind of support because we were sending out a good message of people working together, cooperating, getting things done, and putting education and technology shows in tent after tent we set up. So it was obvious to them that we weren't just rowdy concertgoers, but sort of good guys. In fact, the sheriffs were so behind us they even gave me an honorary sheriff's badge.

We started contracting with companies that put up sound systems and stages and artwork. We also had the most incredible sound system ever done. Not only did we have speakers at the main stage, but we also had extra speakers deep into the audience. This meant the sound in the back was delayed exactly to the point where it would match the front ones. So everyone could hear the music at the same time.

We also had groups to set up lots of concessions. We set up a technology fair with companies like Apple in air-conditioned tents, where they could show off computers and other products. We even had carnival rides planned. I ended up paying a total of about $10 million to complete that amphitheater. That was the biggest expense.

There were also very high payments to the artists to get exclusives for all of Southern California for that year—so bands we signed like Oingo Boingo and Fleetwood Mac, for instance, couldn't play anywhere else in Southern California that summer.

What I'm trying to get to is this: if I compare the US Festival to starting Apple, there's a huge difference. With Apple, I designed those computers alone. I could make every decision by myself and there were very few little changes and trade-offs. It was like I had total autonomy and total control, and that's how I was able to make everything work.

But with the US Festival, I had to deal with all kinds of people and lawyers. And let me tell you, in my experience, the music industry is the worst of all. And then I had to deal with all the construction and costs and funding. Everybody was trying to make bucks off this. So the US Festivals were a much larger business to start than when I designed computers. In fact, it was the opposite. It was much better funded, it had many more people, and it was a trial, a real trial, from the start.

And I was the only one writing checks. This was my show, from that standpoint. But I felt that in booking groups, I just didn't have the experience. And none of my people did, either. They knew how to organize a company but not book groups. I talked to the concert promoter Bill Graham and signed him up. Now, if you've heard any of the legends surrounding Bill Graham, you know that he normally likes to run the whole show. But he'd been in Europe with the Rolling Stones, and we'd already been doing the engineering, coming up with what the stage would look like, the signs, the companies that would be hired, the sound system, the video. It was the first time ever that a big Diamond Vision display would be used at a concert in the United States.

But Bill had some definite ideas. For one thing, he totally nixed my progressive country idea, and he pretty much laid it out like this: You can't have that kind of music. He said, "If you want the kinds of numbers of people you're after, it's going to have to be a modern rock concert." If I really must, he said, I could add some country in.

He also said you have to have what kids in the high schools are listening to. So I actually went to some high schools and talked to kids. And when they threw out lists of the groups they wanted, all they were doing was relaying what the radio and MTV were playing. It was like all they wanted was two performers: Bruce Springsteen and Men at Work. It wasn't as if they had any special knowledge we didn't have. That was disappointing.

But we put the US Festival together anyway, and soon we were there. In 1982, over the Labor Day weekend. Candi was almost nine months pregnant, and we rented a house overlooking this huge venue. I mean, it was kind of scary to look down one day and see the hugest crowd down there. But we were going to pull it off, I knew it.

And we did, we really did. Though I lost money, that was not the biggest thing. The biggest thing was that people had a good time—and that facilities like the food stalls and bathrooms worked without a hitch. It was over 105 degrees that summer, and we set up a huge row of sprinklers people could run through all day to keep cool.

I still get emails and letters from people who say it was the greatest concert event of their lives. I just wanted everyone to smile, and I think everyone did. And we had a lot of firsts, that's for sure. We were the first non-charity concert ever of that size. We were the first to combine music and technology. We were the first to use that huge Diamond Vision video screen to bring the concert to people sitting way in the back, as well as to people at home watching on MTV, and we also had a satellite space bridge connecting our concert to some musicians in the then Soviet Union. We had Buzz Aldrin, the astronaut, involved in the space bridge, too, and we had him talking to a cosmonaut!

This was still during the Cold War. Back then, people in the Soviet Union, mainly Russians, were much more feared than Al Qaeda is today. The fear at the time was that the communist regime of the USSR would annihilate us with their weapons. Some of our UNUSON group had peace-oriented contacts with people in the USSR, though, including technicians who proposed the first-ever satellite linkup (space bridge) between the two countries.

I liked being the first at things—I always have—so I approved this instantly. Here's how we decided it would work: we would transmit live shows from our stage to a group in Russia. They would transmit a live show back to us on the Diamond Vision. The key to making it possible was that before the U.S. pulled out of the 1980 Olympics in Moscow, NBC had left a lot of satellite equipment behind. So all that equipment was still in a warehouse in Moscow.

Our technician friends in the USSR pulled this equipment out of its boxes and set up a satellite link on the specified date of the US Festival. There was no way we could know if it would even work. Back then, it took two weeks sometimes just to get a phone call into the USSR. We had to get the president of GTE to approve a constant phone call on the date of the transmission just so parties in both countries could talk to each other and make sure it was working.

On the date of the transmission, we weren't even sure it would work. Right up to the second their transmission appeared on our screen—the first day of the US Festival—we weren't sure. But then it came up.

Bill Graham was supposed to announce what was happening to the giant crowd. But he didn't. I ran across the stage to where Bill was viewing some TV monitors and told him to announce it.

Me and the USSR

Doing the satellite bridge to the Soviet Union at the US Festival led me to devote more than a million dollars over the next ten years to U.S./USSR peace efforts. The idea was personal diplomacy. I tried to get normal people, not officials, from each country to meet each other.

In 1988, on July 4, I sponsored the first big stadium concert in the USSR, just outside Moscow, with major Soviet and U.S. groups on the stage. The U.S. groups included the Doobie Brothers, James Taylor, Santana, and Bonnie Raitt. I found a cheap $25 guitar at a store in Russia, and got all the groups to sign it. I still have it. That concert was at the end of a great peace march there.

For doing things like the first space bridges between the US Festival and the USSR and this concert, I became pretty well known in the USSR. But you know what? The U.S. press didn't care one whit. There was almost no coverage.

In 1990 I sponsored two-week trips for 240 regular people—teachers, for instance—to tour the U.S. and stay in the homes of Rotary Club members here.

So I had done the first three space bridges in the Soviet Union. Somewhere around this time, maybe 1989, ABC put on a national TV show purporting to be the first space bridge ever. I actually paid for the connections of this hookup, but ABC never even mentioned my name and took credit for being first. Actually, they were fourth!

But he was certain that the Soviet signal was a hoax and coming from a studio in Southern California. He said, "No way would the Soviets permit a link like this."

But I knew the truth. So I went to the microphone and announced to the crowd that this was a historic transmission from Russia. There was some booing—remember, they were Cold War Enemy No. 1—but I knew we were making history.

To the USSR, we transmitted Eddie Money. They loved it.

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The US Festival was also the first huge concert where anyone got to hear me sing! Have I mentioned I have the voice of an angel? I got up and sang with Jerry Jeff Walker, the singer known for the 1960s song "Mr. Bojangles." The song we sang was "Up Against the Wall You Redneck Mother." Good thing they didn't give me a microphone! Walker was actually the only country guy we ended up getting that year. Remember, I originally wanted the whole concert to be country.

I also got to meet some of the other musicians! I was sticking around with my new baby, Jesse; I mostly avoided meeting the celebrities. I did meet Chrissie Hynde of the Pretenders—she had a baby, an infant, with her, too. And I remember how Jackson Browne came up and introduced himself to me. I was nearly tongue-tied—I was pretty intimidated talking to such a great performer.

The main thing for me was the audience.

I remember riding around with my friend Dan Sokol on little scooters and being just blown away by how much fun people were having.

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And I was exhausted. I'd been up practically all of the last two nights, because Jesse was being born. He was two weeks early! It was September 1, two days before the concert began, we'd just finished the sound check, and at about 2 a.m. Candi woke up with labor pains. So, yikes, we hadn't made any plans for the delivery, none at all.

I mean, we'd been taking birthing lessons and all that up in Northern California. I called the midwife, and she recommended a natural birthing center over in Culver City, which was more than an hour and a half away. We borrowed one of the cars at the house we were renting and drove to the birthing center. But we didn't tell anyone.

I'm sure that the next morning, the morning before the day of the concert, everyone was wondering where the heck I was. But it wasn't until that afternoon that Jesse was born. He was a beautiful baby.

When Candi and I were discussing what to name the baby, I'd gotten the idea that we might have trouble agreeing on a name. I proposed a simple, conflict-free solution: if it was a boy, I'd name it; and if it was a girl, she'd name it. Candi thought this was fine. So when the baby was born, I named him Jesse, a name I'd already planned. First I'd thought Jesse James, but then I settled on Jesse John.

The name Jesse sounded funny with Wozniak, though. So I decided that if the baby was a boy, I would name him Jesse John Clark. So when the baby came out, I exclaimed loudly, "It's a boy!" But no, it was the umbilical cord I was seeing.

But then it turned out the baby really was a boy, and I simply announced, "Jesse John Clark."

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I was so tired, walking around the concert when it started, and there was a doctor who kept injecting me with something to keep me up—vitamins, he said. But I had to do all these interviews—one with Peter Jennings, for instance, and one with Sting beside me. And they were asking me questions about this enormous crowd, and I just did a horrible job because I was so tired.

But there is a wonderful picture—my favorite picture ever. It's a picture showing the moment when I got up on stage on the first day of the concert with one-day-old Jesse in my arms. I told everyone that this was the birth of something great. I meant Jesse, of course, but also the concert. People went wild, cheering and everything.

I will never forget that moment.

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I loved that first US Festival concert, and I knew I'd made so many people happy doing it. We thought from press reports that enough people—nearly half a million—had shown up. So we thought that would make us money. But we lost money, nearly $12 million, because it turned out we didn't sell as many tickets as there were people.

A Big 8 accounting firm we hired explained to me that the reason was that people had been sneaking in. And I believed them.

So I decided to do it again. I said to everyone involved, "Let's do another show. We got such great publicity the first time around. We're hot, it's a sure go." And it was hot. So I thought, This time we'll just have to have supertight controls and make sure everybody has a ticket.

This time, in 1983, we did it over Memorial Day weekend. (We had a country music day the following Saturday.) This time we tried to stick with more of the new-wave music at the time—the alternative stuff. We had the Clash, Men at Work, Oingo Boingo, the Stray Cats, INXS, and a bunch of other bands. That was the first day. And on the second day, we did heavy-metal day.

We did the Soviet satellite link again. We had two more space bridges with the USSR. But we didn't transmit music shows this time. Instead we transmitted groups of us in tents speaking to groups of them, person to person. U.S. astronauts and Soviet cosmonauts were involved, too. It was a very big deal. What struck me emotionally was how similar our values were. Those exchanges dissolved forever in me the effects of a lifetime of propaganda about the Soviet people being our enemies.

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But even though we were more careful in counting the tickets this time, we still lost money.

Another $12 million! You know, I was overpaying bands like crazy. I mean, with Van Halen, I paid a million and a half for its one appearance. I later heard that was the single highest amount paid for a band. And David Lee Roth, though he was nice and cordial when I met him, was practically falling down onstage. He was so drunk, slurring and forgetting lyrics and everything.

But this time we had installed very tight controls, collecting ticket stubs and keeping them. We had turnstiles to count everyone who came in. We also had aerial photographers to get an accurate head count. Plus we knew how many tickets we had sold, making sure people didn't slip in like the last time.

But it turned out that the Big 8 accounting firm was full of crock. The problem hadn't been that people were getting in for free. It was that press estimates of attendance were greatly exaggerated. Both times. So we lost the money because not as many people came as we thought. We didn't sell enough tickets to cover costs.

Still, I think of the US Festival as the biggest, hugest success. I'd do it again in a minute, I really would. It was a tremendous experience for me. Everyone had fun! Smiles everywhere. But on the economic side, well, not so hot. I lost a lot of money, and that was a big disappointment.

One of the most memorable moments for me was when concert promoter Bill Graham came up to me near the end of the concert the first year. There was a huge full moon, and Sting and the Police were onstage. And Bill put his arm around me and said, "Look at this, Steve, just look at it. You're not going to see this but once in a decade. This is so rare."

He told me that afterward, everybody was going to be doing these US Festivals because it was so popular, so fun, and so rare.

Later on, he was right in a way, there were all these huge con-

Paranoia?

On my first trip to the USSR, I decided to bring a number of friends with me.

One afternoon my friend Dan Sokol was trying to take a nap, but he was bothered by some Russian music in his room. I guess Dan was too tired to find the little music knob near the door. That's all you needed to turn the in-room music down.

Instead he propped open a ceiling tile near where the sound was coming from. He saw some wires and yanked them hard. They came loose but the music continued. So Dan got up on a chair and found another speaker in the ceiling. He yanked the wire off that one but the sound continued. He probed until he found another speaker, part of an intercom.

Hey, he thought, this is how they listen to you! When he ripped that one out, the sound stopped. Dan took credit for finding the USSR surveillance system. Like they were spying on him. Ha. I laughed because I thought, Well, that's Dan for you, paranoid and into conspiracy theories.

We told this story about the Russian surveillance device to some of our friends who went to the USSR later. The next year, a friend of Jim Valentine's went to St. Petersburg to install some sound equipment in a disco. Thinking of Dan's story, he scoured his room looking for the hidden surveillance device Dan had described. Under the rug he found some lumps. He lifted the rug and saw a brass plate secured by four large screws. He undid the four screws with a screwdriver.

When the last screw came out, a chandelier crashed on the floor below.

Also, around this time I met a girl (I was separated from Candi by this time), a Russian girl—Masha. She was to become a long-distance girlfriend for the next half year. She was an interpreter.

In Russia, my friends would point out several signs that I myself was being "watched." They thought certain Russian officials—car drivers and the like—were KGB agents staying extra close by me at all times.

One time, to get some time alone with Masha, I actually pulled a stunt to ditch the concert in a way that might lose the people who might be tailing me. So instead of leaving the concert in my own Soviet-provided car, I got someone else's driver to take Masha and me back to my hotel, where we had about twenty minutes alone to talk.

The next day, Masha and I toured an art museum at the Kremlin. Inside, she told me matter-of-factly, without even a raised eyebrow, that I was being followed by the KGB. I pooh-poohed this, but Masha pointed to a youngish man in a nice suit standing in the hall we were in. She said, "He's KGB."

She said she could always identify the KGB because she knew a bunch of guys in the KGB school, and she could always spot them by the way they stood and the way they looked. I decided to call Masha's bluff. I said, "You mean, if we backtrack through a couple of halls, he'll follow us?" She said, matter-of-factly and with total confidence, "Yes."

So we went back through a couple of rooms, and we were talking about things and admiring an icon on the wall when I glanced sideways. And there he was. The same guy, across the hall, looking into a glass enclosure.

I lost that bet.

certs: Live Aid, Farm Aid, all of those. They were concerts in stadiums, though, that were all in prebuilt places. Who else in history ever went out and actually built a facility like this, really a pretty good facility to support and maintain that many people?

For them and for me, it was the highlight of my life. Making money, losing money, that's important. But putting on a fine show is most important of all!

Chapter 18

Leaving Apple, Moving to Cloud Nine

After the US Festivals and graduating from Berkeley, I went back to Apple to work as an engineer again. I didn't want to manage people or be an executive or anything like that. I wanted just to be there and design new circuits and come up with clever ideas and apply them.

But once I got there, it was weird, because I was already in the media mainstream and had so much other stuff to do. A ton of stuff. I was being called by a lot of people—the press, computer groups I had to speak to—and I was working on these philanthropic projects like the San Jose Ballet and a local computer museum. I was sort of splattered all over the world and in all these countries and in all these different areas—beyond just working on circuits.

So I could get the engineering sort of started and come up with an architecture idea. For instance, it could be something that might speed up the processor five times, but the other engineers had to do most of the actual designing of the chips and the connections and the laying out of the printed circuit boards. So truly I felt it wasn't critical for me to be there, even though I loved Apple still.

I was working in the Apple II division. This was after the Apple III project was closed down, so the engineers from that department became Apple II engineers. A lot of them just gravitated around me. It was fun. And there were some cool people with some cool projects starting up in my building at the time. For instance, just as I got there, on the next floor down from me, they were finishing up the Apple II C computer. This was the small Apple II—a really small one—as small as today's laptops except you had to plug it into a wall. I thought it was just a beautiful computer, my favorite one to this day. I really think it was one of the best projects ever done at Apple.

Well, one of the engineers on that project was this guy Joe Ennis. Joe was the kind of guy I love, the kind who's so enthusiastic and passionate about the products he's working on and where they can go and what he could do with them. He had long hair; he was kind of hippie-looking even though it was already 1985. And he had these ideas—all sorts of great ideas about extending the Apple II into areas far beyond anything even the Macintosh people were talking about.

Like he thought you could have an Apple II programmed to be a complete telephone switchboard. (Today, telephone switchboards are just computer cards you plug into a computer.) He imagined you could store voices digitally—that was so ahead of his time—and he thought you could route them out to other channels digitally. He just had idea after idea after idea about the future of computers. I thought his brain and his ideas were just wonderful.

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Now, for a while I'd had this really nice home up in the Santa Cruz Mountains with all this really high-end audio and video equipment. By that time TV sets all had remote controls, and VCRs did, too. I also got into laser discs, so I had a remote control for that, too. Then there was this expensive hi-fi system I had from Bang & Olufsen. It also had a remote. That was rare at a time when no other stereos came with remotes.

I was also, at the time, considering getting a satellite TV. This was so rare; you couldn't even buy them in stores. I lucked into mine through a friend, Chuck Colby, who was building custom dishes for people. But, man, there it was. Another remote control.

So typically I would turn on the TV with one remote control and maybe I'd turn on the hi-fi with another (because I had the speakers routed to the TV), and then I would turn on the satellite and then I'd press a few buttons for the channel I wanted on the satellite and I think I had to turn on my VCR to pass a signal through it—all the signals passed through it to get to the TV the way I had it hooked up. I'm pushing all these buttons on different remote controls and it was just obvious to me.

Here I am, sitting in bed, operating all of this equipment with all of these different remote controls. It was crazy. I wanted one remote control with one button that was programmable to deal with all of the devices. I didn't want a button that turned on a TV and another button that turned on the VCR and another button that turned on the satellite and another button that selected the satellite channels and another button that entered that number.

I wanted one remote. Just one. And I wanted that one main button to be able to do multiple things. I wanted to push it and have it go zip, zip, zip, zip, zip and have all the infrared signals come out of one remote control that turned everything on to the status I wanted.

If I wanted to watch a laser disc, it'd turn on the TV, then select input 3 on the TV, and turn on the laser disc player and start playing, for example.

So it was real clear to me that a single remote control solution was necessary. And I knew I was able to see it before most people, because most people in the U.S. at the time didn't have as many different remote controls as I did. Most people would look at me and say, "What do you mean? I only need two remote controls, one for the VCR and one for the TV."

But I realized that soon people would need more remote controls and it would become a problem, like it had already become for me.

I started talking to a few people about this idea and I got excited because I realized how easy it would be to build. This would actually be an easy project. A little microprocessor can look at the codes coming in, store the data, and then output the same codes when you press the buttons.

And I like to be first, as you know. And I thought, I'm the one to do it. And I really did become the first person in the world to do what is now known as a universal remote control.

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Let me go a bit deeper and explain exactly what it was that the remote, as I designed it, was doing.

As I said, it was very important for me to make sure this remote didn't have to have one button corresponding to each button on each corresponding remote control. In that case, I would've had a million buttons—all the ones on the TV remote control, plus all the ones on the VCR remote, plus the ones for the satellite TV remote, and so on.

I wanted a *single* button on my control to sequentially emit many infrared codes corresponding to buttons on another remote control—or even countless other remote controls. I didn't want to, as a consumer, have to press five buttons in a row just to turn things on and turn it to my favorite starting channel—in those days, The Movie Channel. I wanted to press one button one time to do all of that.

That meant that the buttons on my control were like macros. One button could represent a whole sequence of things. (In Word, for instance, you could set up a macro on one key so you could just hit that key—say, CTRL+S—to check spelling on your document, accept all changes, and then save the document again.)

I realized that this is exactly like a program. I would have to write a little program for each button. So I came up with the idea not only to let the consumer decide what a button does but also to reprogram the button to redefine what another button does. I built a programming language into the remote, and I went a step further, adding an ability, referred to by the prefix "meta," that would allow a program on a particular button to write an entirely new program for that very button, for itself.

It was a beautiful language, I was proud of it. As it turned out, it wasn't the easiest way to do what the vast majority of users would want, but it would be very attractive to software geeks like me.

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I was still at Apple when I got this idea. And I started telling people about it. People like Joe Ennis. As I said, I loved the way his brain worked. He was always interested in unusual uses for technology. And I told him about my remote control idea and we started talking about it all the time. He really got it.

So I pitched real heavily to Joe the whole idea of, "Let's leave Apple and start this company."

I never felt like I was turning my back on my own company. Never. By this point Apple was a large company, and it wasn't and still isn't the love in my life. The love in my life is starting small companies with small groups of friends. Bringing new ideas out and trying to build them. By then the Apple idea wasn't so new.

At the time, I was heading up a new Apple II that was supposed to be better than anything, called the Apple II X. But shortly after we started, upper Apple management canned it.

Looking back, that was probably a decent decision. After all, they were used to products that sold 20,000 a month, and a high-end product like the Apple II X, because it would be so expensive, probably wouldn't sell more than 2,000 a month. So, like I said, they canned it.

Another Apple II product was actually born out of the Apple II

X: the Apple II GS. The joke was that it stood for Granny Smith, a type of apple, but it actually stood for graphics and sound. And that was a great project. With graphics—real graphics in 24-bit color that worked with computer monitors instead of TVs—and sound—real sound, not just chirps—suddenly you could do really interesting things. Like games and software for kids, who really would need that level of production to be engaged.

I was so happy to see that we got a project that all of a sudden brought the Apple II into where it really needed to be. There were some morale problems in my group as a result of the people in the Apple II group feeling undervalued compared with the Macintosh group. (The Mac was in development.)

And I was ready for something new.

Very quickly after I started talking to Joe, and also to my assistant Laura Roebuck, I decided I was going to go ahead and do it—start a company to build the remote control. They both wanted to do it. And I was so lucky to get Laura—she'd just had a baby and wanted to work part-time, and Apple didn't have part-time positions.

Anyway, it was such a simple idea; I really didn't need a bunch more engineers than Joe and me. (Things are different now, of course. A venture capitalist would make you hire twenty right away!) But this was in February of 1985.

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The first thing I did was to call my boss's boss, Wayne Rosing in the Apple II division, and tell him I was leaving to start a remote control company. You know, I had a job and had to tell someone, "I'm leaving. I'm leaving to start a company."

I didn't call Steve or Mike Markkula or anyone on the board. I had a job in engineering, and I felt like I just had to tell someone I reported to so they would know.

I sat them down and sketched out my idea and described it just like I've described it to you. I told them I was doing a remote control that would be a single remote control that would work with all of the consumer electronics someone had. It was going to be one remote with one button, very simple. It would not compete with anything Apple did.

They gave me a release very quickly, saying they'd seen my design and there was nothing competitive about it. The letter also wished me well.

I left within about a week, but I did stay on the payroll as an Apple employee. I am to this day. I just have the absolute lowest salary a full-time employee can have. I still represent Apple at computer clubs this way.

Steve probably heard I was leaving the same day almost everyone in the world heard it—instantly—the day a piece came out in the *Wall Street Journal*. But the piece got it all wrong.

The reporter called me the very day I was leaving, the day I was packing up, and said, "I understand you're starting a new company?" So the rumor was out. I told him yeah, and he asked me what it was all about. And I told him.

He asked me, "Are there any things that you aren't happy about at Apple?" And I told him the truth. I told him yes, and then I stood up for the people I was working with who were offended by the lack of respect they received.

At the time I was leaving, the people in the Apple II group were being treated as very unimportant by the rest of the company. This despite the fact that the Apple II was by far the largest-selling product in our company for ages, and would be for years to come. It had only just recently been overtaken as number one in the world by the IBM PC, which had connections in the business world that we didn't have.

If you worked in the Apple II division, you couldn't get the money you needed or the parts you needed in the same way you could if you worked in, say, the new Macintosh division. I thought that wasn't fair.

It boiled down to certain kinds of expenses, what kinds of components you were allowed to buy from other companies, how much money you were allocated to work on projects, despite having such a hugely successful computer in the world. Like I said, a lot of things were being trimmed way down.

Also, there were limitations on the Apple II in terms of taking advantage of new advances in technology. We'd hear, "No, Apple II will stay the Apple II, and we're not going to let it move into newer, more advanced areas." Things like that.

So I made some comments like this, and then the reporter asked, "So that's the reason you're leaving?"

And I said, point-blank, "Oh no, that's not the reason. I'm leaving because I want to do this remote control."

But the *Wall Street Journal* printed the article suggesting I was mad at Apple and that was the reason I was leaving. It was wrong, very wrong, because I went out of my way to tell the reporter not to get it confused. Maybe it was more interesting to shape the story the way they did. They just left out a couple of words, the words "That's not why I'm leaving," and that was the same as implying that that was the reason I was leaving.

Oh my god. I have to think it was an accident, but let me tell you. It's been picked up by every book and every bit of history ever since. It's just wrong. I mean, they asked specifically, "Is that the reason you're leaving?" And I went out of my way to say "No." But it didn't make it into print like that. Everyone in the world ended up thinking I left because I was mad at Apple or something.

The only reason I left my day-to-day-job at Apple is that I was enthusiastic about the idea of doing this new neat project that had never been done before. I saw that remote controls were going to be more important in people's lives as satellite TVs and other devices came in. Remember, there was no store you could go down to yet to buy a satellite TV. You had to be in a select group of people to know how to even buy a home-built receiver for that.

If I hadn't had the remote control idea, I would have stayed right where I was. But this was such a cool idea. And we got moving pretty quickly.

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Our first thoughts were where to locate. I lived on Summit Road in the Santa Cruz Mountains. Up there at the Summit were two restaurants, the Summit Inn and the Cloud 9. I knew that the Cloud 9 was closing, so I suggested it as a site. How cool would that be?

Joe Ennis picked up on the name Cloud 9, too. We had the lawyers who incorporated us check it out, and they found that Cloud 9 was taken. I can't remember which of us came up with CL 9. It may have been that I saw it on a license plate, but it's hard to remember. At any rate, we settled on CL 9, which was still a great name.

Maybe two weeks after that, we got an office in this older part of Los Gatos, the town where I lived. It was right up against the Santa Cruz Mountains on the last little corner with a few shopping places. The space was small—maybe around 900 square feet—and it was right above a Swensen's ice cream place. And that's where Joe, Laura, and I moved in.

It was great. It felt just like the early Apple days, so exciting. We were building something no one else had thought of yet. Who had ever thought of playing your remote control into a device that would learn its code? That had never happened. I mean, today it's more obvious because we have universal remotes and things, but not back then.

The first thing we did was to start meeting with representatives of components that might apply: infrared sensors; infrared transmitters; microprocessors. We started looking over the data sheets and the books and figuring out what microprocessor we'd use. We started making some choices and got together an idea that was more in our heads than on paper, but it wasn't like a finished design you could actually breadboard and wire together and build something from. This process was exactly like what I did with the Apple II.

There were a couple of areas that were tougher. One problem we had to tackle was how do you receive infrared signals in the remote? We didn't have that much expertise in this area. I didn't, and Joe wasn't so sound on how you build a sensor for infrared. So we actually hired a consulting firm in Sunnyvale to help us basically read an infrared signal. If you placed your own remote control right next to our receiver, the signal coming out of yours was a very strong signal.

In the same way that the closer you get to a lightbulb the brighter it is, so it is with a remote control. The consultants designed an intricate circuit with an awful lot of parts and filters. And I said, "If you're close and it's powerful, then why can't you just detect it with a lot simpler circuitry?" Just go straight to a photo transistor. You know me. I like to do things with the simplest possible circuitry. You don't need all these special amplifiers that need power all the time. Just go straight to the photo transistor, which is a transistor that senses light instead of electronic signals.

And that idea actually worked out.

They had to put in a couple of little parts and capacitors to filter it just right to avoid the signal bouncing around in a weird way. And they came up with a very good circuit that worked reliably. You could play your remote control into our little receiver device and it would capture the signal very accurately. It could determine how many microseconds the infrared signal was on and how many it was off.

Then it could trace the signal and make a time recording of it, the signal from your remote control.

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The time also came for us to come up with a plastics design for the remote. Very early, just after we moved into the second building on Alberto Way, we started going around to a few design companies to see if they would show us some samples, or some ideas we could look at.

One of those companies was Frog Design, which had done the Macintosh. We called them up and they said, "Sure, we do third-party development for people other than Apple."

They said they'd take anyone who walked in the door and talk to them, design for them. We told them what we wanted, and they did it up in a few models. A couple of them were a little too fancy-looking for my taste. I wanted the most normal-looking design, just totally straightforward, where every button is sort of square, a real symmetrical design.

I wanted it to look like an everyperson product, not something that looks like it's from outer space, you know what I mean. And we liked some of the other products they came up with to fit that description.

But in the end, they dropped us.

It turned out Steve Jobs was over at Frog for some reason and saw a CL 9 prototype. From what I heard, he threw it against a wall and put it in a box and said, "Send it to him." As if Apple owned it. The Frog guy told me that Steve told him they couldn't do any work for us because Apple "owned" Frog. Not true, and everyone knew it. But Frog told us they felt uncomfortable doing it without Apple's permission—Apple was a big customer—so they weren't going to do it.

Well, I wasn't going to argue. I don't truly know what the real story was, but I thought, Good, fine. We'll go somewhere else. And we did go somewhere else.

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Of course, I had to choose a microprocessor for the device. I ended up choosing two processors. So the remote later went down in history as the first remote control with a dual processor!

Anyway, thinking about the two microprocessors and working with Joe, I decided it might be nice to have one microprocessor for small tasks like reading the keyboard and keeping time, and another to do the heavy-duty work. The larger processor I used was an updated version of the old MOS 6502 I'd used for the Apple I. The other was a smaller, cheaper processor. I think we paid 50 cents apiece for it in quantity. It was a 4-bit processor—meaning, remember, it could process only 4 bits of data at a time. That was all we needed for these smaller tasks.

However, a little processor like that is hard to write a program for. Man, was it hard to control. It was almost as hard as programming the state machine in the floppy disk. Nothing was built into the hardware, and when you don't have the hardware resources, you have to take advantage of what you do have inside the chip. And you wind up with weird instructions that do things in weird ways. That's because the chip didn't already have built into it well-thought-out instructions a human could easily understand and use. That was to keep the cost to a minimum.

But the program on the 4-bit microprocessor wound up doing two basic things: keeping the time of day, monitoring the keyboard, running the LCD display, and enabling power to some of the rest of the circuit, and it also communicated with the bigger, 8-bit microprocessor, telling it what keys had been pressed and receiving data to display on the screen.

We sat down and sketched out on paper where we wanted the lines of letters, numbers, and a few special words to pop up on our LCD. And we found a company that would make us an LCD. We gave them our layouts, and they eventually brought us back LCDs with a bunch of connection pins. And that LCD would actually connect to the same 4-bit microprocessor chip that was reading the keyboard.

Now, the real guts of our product—memorizing all these different infrared codes and repeating them when you pressed buttons—was going to be done by the second, more powerful microprocessor. Because of the updated version of the 6502, I thought, Great. I am so familiar with this! It had a very beautiful architecture inside. The way it was structured to be, with very few transistors inside doing a lot of work, it was just so good and it did the right job.

The Apple II had my little development system that I wrote myself, and I could type instructions in quickly and test them out. What if I could have that for this microprocessor? So we actually designed our board in such a way that you would be able to hook up through a serial port so that we could connect a terminal or computer directly to it. That would enable you to type and see data on the screen, although the remote control was really the computer. (It was like a little cousin of the Apple II.)

What terminal? Well, I decided the Apple II C would be a great terminal. There were programs that could make it behave like a terminal that talked to other computers.

Remember how I told you that in the Apple II I had added this mini-assembler that let me type in things like LDA for loading the A register, or #35, which meant 00110101, the binary language of 1s and 0s that computers can understand? That program and many other development tools were built into the Apple II, but would really be useful in the remote control as well.

I had a friend I'd worked with at Apple, John Arkley. He was a consultant, and he offered to convert the debugging and other code I'd written for this new 6502 microprocessor. We paid him and he did it.

And it was great. I would hook a little Apple II C into our breadboard—our wired prototypes—and type away and do the debugging. It's like I had a new little Apple II inside of the remote control. It had all the fun of an Apple II.

And when we finished designing it, this product was great. We were all just blown away by how great these tools turned out.

So then we had a manufacturing issue. Who was going to manufacture the device? And suddenly an old friend of mine showed up from my childhood days with the Electronics Kids. Remember my neighbor Bill Werner? He was the one who did all the toilet-papering of the houses with me and got all that phone cord to create the house-to-house intercom system in our neighborhood.

By high school, though, Bill had kind of gone in a bad direction. Not like me. He got bad grades, got a motorcycle, got in trouble for burglarizing an electronics store, got into some bad stuff. But now he'd turned his life around, and we ended up hiring him—he had worked at the Silicon Valley manufacturing firm Selectron. And we hired his wife, Penny, too, to do some secretarial work. So our team was building.

Selectron was the kind of company we needed. It did manufacturing, like I said, and that was the one thing we had left—figuring out how to build this device in mass quantities.

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Meanwhile I got a call one day from a venture capitalist in England. You see, years before, in the early days of Apple before we went public, he'd called me up and offered to buy some of my stock at a low price, and I'd said yes. But he hadn't bought it.

Well, he called at a slightly later date and asked again if I would sell him the Apple stock at that price. I'm not sure what the offer was, but it was low. By this time Apple stock was easily worth ten times whatever he was offering, even though it wasn't yet public. He said, "You promised to sell me some at this price. Will you?"

So I kept my word. His venture capital company made a ton of money on the London market, a ton of money.

Now, at CL 9, I told him all about this new company I was starting, and he said, "Can I visit you?" I told him sure. He showed up. I remember thinking, Man, this guy is really staid. Just very formal—so reserved in his language and manner. He was English, okay. I guess he was stuffy compared to us, and you can imagine how loose we were.

Anyway, I described to him what we were doing, and he immediately said he wanted to invest. I told him I wasn't taking any money, that I was financing it all. But he actually begged me.

Well, when people beg me and say they want to be a part of something, I always give in.

After his investment came in, I suddenly had another big investment from the big Silicon Valley venture firm New Enterprise Associates (NEA), which had also done 3Com, Adaptec, and Silicon Graphics. This guy from England had brought his friends in, you see. So all of a sudden we had two or three million dollars.

So we pulled that off in a few months, and we began to realize we were going to need a bigger place to set up shop. I called an old friend of mine from Commodore, Sam Bernstein, a guy who'd written articles for newspapers. He was sort of a reporter. And I always liked the way he thought and the way he organized his thinking. So I asked him to come on board—this was early on—as president. We got along splendidly.

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We ended up keeping CL 9 in business for about three years, maybe a little more. There are still people out there who talk about how amazing our product was. I don't regret doing it for a second. I ended up selling the company to someone, but they couldn't raise money and closed it down.

But at the time, I had other challenges to think about. I had two small children at home (Jesse and Sara). So it was hard making sure I had enough time to devote to them.

I mean, after the 4-bit microprocessor project was done, it was time to do the 8-bit. And I set out to do it and was just having a lot of difficulty getting started on that job. I had my kids I was giving a lot of attention to. And my relationship with Candi was starting to get rocky. We were fighting. We weren't getting along at all. We had fights about how to raise the children, especially. And we were talking separation.

Well, I had an idea to just take off and hang out in a hotel room somewhere beautiful for a week. I planned to just disappear from the world and go to Hawaii and write the code.

So I went to Hawaii, the Hyatt on Kaanapali Beach, and I set up my little Apple II C so I could start typing the new program in. (Someone was watching the kids.) I thought solitude would help me finish the project. At least I hoped so.

But what happened was, I didn't do a single thing that entire week. I literally sat there looking out my window and watching whales every day; I got used to the hotel schedule. I swear, about ten times a day somebody would come into the room to restock the minibar, change the sheets and towels, check this, check that. All day there were these major interruptions. I hated that.

So after that week of doing nothing, I thought I should stay another week. I found out I could keep that same room that I loved for another week.

Well, guess what? I wound up staying there for four weeks and not doing one single bit of code. I did nothing there, absolutely nothing. I just enjoyed being there. While I was there the *Challenger* space shuttle disaster happened—it was January 28, 1986—and that was really extremely upsetting to me. But whatever the reason, I did nothing.

At first I thought this was okay. Many times in the past, as I've described to you, my head is thinking about a problem ahead—it's all in my head—and by the time I sit down to write things down, the code, I can write it really quickly and productively. I can do a lot in a short time because I've figured it all out beforehand. So I expected that to happen and it didn't.

It was then that I thought, You know what? There are a lot of engineers in the world and I've got kids. I think I'd rather just hire somebody to do this part of the code. It was like I had sort of reached my limit of being able to mentally—with the 4-bit microprocessor—put myself through this kind of design effort.

So we hired another programmer to do that job on the 8-bit microprocessor. I wanted to spend more time with my children.

I stayed at CL 9 for another year, but that was really when my life changed once again.

Giving It Away

I didn't start Apple so that I would get more money than I would ever need to live on. I never planned in my life to seek great wealth. And I'd always been inspired by stories of those who gave in order to do good things in life.

So I felt this was the right thing to do. And it felt good. I was around people on the boards of the museums and the ballet who were more inclined to social activity. They were less about humor and jokes, less than I was, anyway. But they were good people who believed in what they were doing, and I believed in them.

The first project I funded was the Children's Discovery Museum of San Jose. I funded it entirely for many years, eventually to the tune of a few million.

Then I helped start The Tech of Silicon Valley, a computer museum. I also did the initial financing for the San Jose Cleveland Ballet, now known as the Ballet of Silicon Valley. Why ballet? Again, it was the people. They were great and I had confidence in them.

I also contributed to an expansion of the Center for the Performing Arts in San Jose, which benefited both the ballet and the orchestra. This was a donation that would directly benefit the city of San Jose. How neat to donate to a city.

And though I didn't expect it, in 1988 San Jose's mayor, Tom McEnery, called me to say they were going to name a street after me! In fact, it would be the same street the Children's Discovery Museum would be on. The name of the street is Woz

Way. And it's one of the proudest things in my life—to have a street named after me! Not a dumb name, but a cool name. It would be a bummer to have a dumb-sounding street named after you.

Chapter 19

The Mad Hatter

I think there's a time in everyone's life when you look back and ask yourself, What else could I have been? What else could I have done? With me there's just no question about the answer, none at all.

If I couldn't have been an engineer, I would've been a teacher. Not a high school teacher, not a college teacher. A fifth-grade teacher. I specifically wanted to be a fifth-grade teacher ever since I was in fifth grade.

This was something I wanted to do since so early in life. Who knows where these things come from? Probably because my fourth- and fifth-grade teacher, Miss Skrak, was so good to me and I liked her so much. I felt she had helped me so much in life by encouraging me. And I believed, truly believed, that education was important.

I remember my father telling me way back then that it was education that would lift me up to where I wanted to go in life, that it could lift people up in values. I remember how he said that the world was kind of screwed up at the time—there was the Cold War between the USSR and the United States and all that. And he said that with education, the newer generation could learn from the mistakes of their parents and do a better job.

I felt these were really mighty goals in life: looking consciously at the sort of person you want to be, the sort of life you want to live, the sort of society you want to help build.

But by the time I was in high school and college, I'd kind of forgotten about my goals of working in education. There were times when it glimmered back at me. This girl at Berkeley, Holly, the first girl I kissed, well, a relative of her roommate brought around to our dorm a baby, four months old. And Holly, who was interested in child psychology, started doing all kinds of little games with the baby, trying to test where the baby was in its own head. Like she'd move a pencil and see if the baby's eyes would follow it, that sort of thing. I remember how that just struck me that day, this notion of cognitive development. How shocking it was to me to suddenly realize that the mind really develops in identifiable stages. Almost like logic in a computer, it's predictable. It was like logic, the thing I was into at the time, an intriguing kind of process—a game with rules.

That made me really remember my desire to be a teacher, and for the rest of my life I was always paying a lot of attention to children wherever I went. Infants, babies, younger children, older children. I'd try to relate to them, to smile, to tell them jokes, to be kind of part of their company. I'd been brought up with the idea that there were "bad people" who might hurt children or kidnap them, so I decided I would be a "good guy" any kid who met me could rely on.

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Some people just love being around children, others don't as much. I remember one summer when I was working at HP, Steve Jobs told me he really needed a job for some extra money. I drove him down to see the job listings over at De Anza Community College, and we found this job listing for people to stand in Westgate Mall for a week dressed in *Alice in Wonderland* costumes. They needed an Alice, a White Rabbit, and a Mad Hatter. I was so intrigued. I drove Steve down to the guy who was interviewing people and telling them what it was like. Basically you put on these costumes, he said, and carry some helium balloons and you stand around. You can't talk to the children, but they'll all be around looking at you, he said.

"Can I do it, too?" I asked. I loved the idea. So basically, they hired Steve, his girlfriend Chris Ann, and me as the *Alice in Wonderland* characters. We took turns in the costumes with some other people because, even after a twenty-minute stretch, these costumes got terribly hot and sweaty inside. You could hardly breathe. So sometimes I would be the White Rabbit and Steve would be the Mad Hatter, and sometimes it would be the other way around.

It was kind of funny because you had really limited mobility in those big costumes. I remember I went out as the Mad Hatter once, and all of a sudden about ten kids started grabbing me by my arms and my sleeves and spinning me around. For fun. They were laughing! And I couldn't say anything to stop them, because there were a lot of kids doing it and I wasn't allowed to talk. They could have toppled me! I was lucky they didn't.

I thought this job was so fun I even cut back my engineering hours and took an hourly minimum wage for that week so I could spend more time doing it. I loved looking at the kids' faces when they saw us. I just loved it.

We'd take lunch breaks in our regular clothes and eat at this little restaurant in the mall. One day this little kid—this tiny little kid—points at my tennis shoes and says, "Hey, he's the Mad Hatter!" I told him, "Hey, be quiet!" Ha. That was a very fun week. So fun.

But Steve didn't enjoy it as much as I did. I remember years and years later, I was commenting to him how much fun that *Alice in Wonderland* mall job was, and he said, "No, it was lousy. We hardly got paid anything for it." So he had bad memories of it, but

I just had the best memories of it. I guess I thought everyone was like me and would like doing something like that with kids.

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I loved being a parent, too. It was great. I didn't read books on parenting; I didn't want to read about any structured rules. I wanted to relate to and communicate with the child. Because if you can talk to them, then they'll talk to you about most of the things in their life. I wanted to expose them to creative thinking, I wanted to show them that you don't have to narrow and restrict your thinking the way so many people do. I never once tried to impress even my own values in life on any one of my kids.

I wanted to be like my dad. I remember his conversations with me; he would always point out all sides of an issue. I would know what he thought about it, but he would let me come to my own decisions, which very often turned out to be like his. He was a very, very good teacher. So I intended to be that way, too.

Candi and I had three children. The first was Jesse, who was born the night before the US Festival that Labor Day weekend in 1982. Then Sara came two years later. And Gary was born in 1987, after Candi and I had already divorced. So that was hard.

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With Jesse, when he just a few months old I had the most fun with him doing what I called these "flying tours." I would hold him so that his belly was over my palm and he could see everything from the correct perspective. (I got the idea from Candi's brother, Peter Clark, who told me that if you hold a baby on its back, it's always seeing everything differently than grown-ups do.) But the other way, the baby could see the world like we do. It was just logical.

So I used to hold baby Jesse that way, and all of a sudden I could see his eyes would look to the left or the right a little. Then his head would move in one direction and stay there, and I'd realize, Oh, okay, he's looking at the window shade. So what I did was

I'd take him over to it. It was only fair. I'd let him touch it—I'd move his hands against it—and when he was done, he'd turn his head again, like maybe back toward his mom, and we'd zoom back to her.

So we started getting in the habit of doing this. He'd be lying on my palm, looking at the big TV, and I'd take him to it. Or to the shelf, which had a top and an edge he could feel. So he started getting around the world this way, and he'd always come back to home base at the end.

Jesse got more and more confident. We'd start from home base and then go room by room through the entire house. He'd explore. I could feel his muscles tense in a certain way I could interpret as "Lift me up a little more" or "Let's go a little lower." Sometimes, when he got a little bigger, he would wave his arms and his feet like he was a mad swimmer, and that meant "Go as fast as you can." So we had this great form of communication between us, and this was all before he was even eight months old. I was no longer just looking at the movements his head made; I'd feel his muscles tensing to tell me which way to go. I used to tell people this, and they didn't believe me. So I'd tell them, "Okay, I'll close my eyes. Drop something." And then Jesse would just tense his muscles and lead me right down to it. It really surprised people.

I would try this with other babies—these flying tours—and I found out that after about twenty minutes, I could do it with them, too. All babies were the same! All babies gave the same muscle signals. I loved that I had figured out a way to let Jesse choose what to explore, before he could even crawl or walk, without having to be totally dependent on someone else.

When Jesse got bigger and too heavy for the flying tours, I got into these little Honda scooters. I had the little 80 and 120 cubic centimeter scooters. They're real small, like a bike with a little motor in it.

Up there in the Santa Cruz Mountains where we lived, there were a lot of little windy roads and very few cars. So I could put Jesse on the scooter and we could just go everywhere. I'd let him decide if we would go left or right, and I'd describe things we saw and then let him touch them—we'd say the words "leaf," or "water," or "tree." He chose every single turn we made. Eventually—over a period of a couple of years—he could get into his favorite routes. I remember these as such wonderful, wonderful days.

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By 1988 I was a full-time dad. I was finished with CL 9. By then we had also had our second child, a girl this time, Sara. Sara and Candi became really bonded, as bonded as Jesse and I were.

But Candi and I still weren't getting along. By this time we were already heading for divorce. A critical point happened the night after a concert at the Shoreline Amphitheater. We had a tradition with Jesse that the front passenger seat was the "story seat" and whoever got to sit there would get a story I would make up from the driver's seat. Now, I'm not a writer, and don't ask me how I did it, but I could come up with the most amazing stories. Science fiction stories, usually, and they would go on and on.

But one night Candi and I got into this fight. She felt like she'd drunk too much to drive, and she wanted me to drive. That was fine with me. But she wanted to sit in the story seat, the front passenger seat. Jesse objected, because he wanted to hear a story. And I begged him, begged him, to please sit in the back and I would still tell a story. But he wouldn't get in the backseat. And Candi and I got in the hugest fight because of that. Very shortly after that, it was divorce time.

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So now, suddenly, I was in a new house of my own in Los Gatos. The kids spent one week at my house and one week with Candi. I didn't have any business going on, CL 9 wasn't going on, so I could focus all my energy on the kids.

It was at about this time that I redirected my philanthropic activities from museums and ballet to schools in Los Gatos. This was about 1989, and computers in schools were starting to become the big talked-about thing. There were going to be computer "haves" and computer "have-nots." So I started providing computers to schools—setting up computer labs with dozens of computers in them as gifts to the schools and the kids.

Eventually I worked out a deal with my local elementary school, the one that Jesse was by then attending. It was the Lexington Elementary School in the Santa Cruz Mountains. It sure was an unusual environment for a school. Not like the ones that are all flat and spread out. It looked rustic, out in the middle of nature with the mountains, the trees, and the Lexington Reservoir nearby. This school was also neat because it only had one classroom for every grade. It was a small school.

So I got to know a lot of people there, especially the school moms, who really are the ones who do so much of the stuff. I couldn't really bring myself to do the home school club—that was the PTA-style group they had—I couldn't make the time. But with the computers, I thought I could make myself useful to the kids and the school in other ways.

Around this same time, I started teaching Jesse about computers. He was in the fourth grade by this time. He would go into his room—he had his own computer with a keyboard—and he was this little kid sitting in front of it. He would just type away all day long. At first he couldn't type very well. He would hunt and peck. But very quickly he learned how to cut and paste text from one page to another so he wouldn't have to retype it.

By the end of fourth grade, his computer skills had evolved so quickly. Sometime during that year, he was actually answering questions I had—like if I was having trouble finding out where something was in the system, he would tell me what menu to look in. I showed him how to do spreadsheets and do calculations so he could do his math homework from school with it. He could set it up and do it all on the spreadsheet so the teachers wouldn't see the formula, just the answer. But, of course, I told him he had to do it by hand first, before doing it on the computer. He had to have one handwritten one done, to show me he knew how to do it, and then he would turn in these really nice printouts.

Believe me, there was no other kid in fourth grade who was turning in a spreadsheet and printout for his math homework assignments.

And Jesse just loved doing it. He always stuck by the rule of having to do the homework the real way first—before you do it on the computer. But he loved any homework he could do with his computer. Like typing reports. He loved that.

That year, one of Jesse's schoolmates, Elena, was having trouble in school. I'd known her since she was born. Her mom called and told me that her grades were going down; she wasn't achieving, just having a really hard time. I cared a lot about Elena. So I decided I would go over to her house and we would sit down together. I would take her through ideas to put in reports she would write. We would try to put in comedy—just to make it fun for her. And I'd show her how you do it on the computer.

That became her motivation, doing it on the computer. It was something special, and she really got into it. All her grades in school started going up. Her parents gave me all this credit. She loved doing any homework she could do on the computer; she was a smiling girl and doing well in school. She grew into a woman who is today an incredibly great speaker as well as an actress.

So then I started thinking. If this was so successful with Elena—taking her from basically flunking out of school to A's and B's—what if I could do that with other kids? Why not give it a try? But I was a little scared. Can I teach a group of kids? What's involved? I really did want to teach them normal things—math, reading, writing, history—but how was I going to be able to do that? I don't have a teaching credential or anything.

So I thought, That's it. I'm going to be a teacher. I'll teach a computer class. The next year was the fifth grade. I took six kids out of Jesse's fifth-grade class and put them in computer class. And we started out the class by unscrewing the computers to look at the parts, and I taught them Base 2—1s and 0s, how numbers were represented in computer language. We didn't carry the Base 2 stuff very far throughout the year; I thought I was going to teach them how computers work. It's something easy for a fifth grader to learn; you don't need higher-level math. And we did that.

But the primary goal was to teach them how to make their homework look good. The state of computers back then was such that this only got about one-third of class time attention. Back then, computers were more unreliable, and more subject to software and hardware bugs. On any given day, a hard disk might stop working. Or a battery might go bad. A buggy program might corrupt some files.

Back then, maintaining a computer was a difficult task. So another third of the class involved maintaining the computers. Installing new software and hardware, identifying hardware problems, identifying and fixing software problems of all types. Finally we spent a lot of time on online and network things. Every single year, from the very first class on, I bought AOL accounts for all the students.

It was important that they learn to communicate with people far away, and in a way that had never been done before. The two things my students did the most were to download fun software like games and freeware utilities and visit chat rooms. I encouraged them to go as far as they could in chat rooms. They found it amusing to pretend to be other people, to pretend to be older than they were. Even though it might take them two minutes to type a short sentence, the girls would claim to be nineteen years old. The boys were always honest.

Some of the girls would get too excited and scream to the rest of their class that they were making a date with another nineteen-year-old! Yeah, right. The thing I always noticed was that the other "nineteen-year-old" was also typing a sentence every two minutes. None of them could type when they started my class, but they sure started learning.

And the things I learned in my ten years as a teacher, well, they're just too numerous to count. I felt this was the most important time in my life.

Chapter 20

Rules to Live By

Maybe you're wondering why I haven't written a memoir before this. People kept asking me to. There are a lot of reasons I didn't. I was busy—too busy. A couple of times I even tried to start working on it, but my plans always fell through. I just didn't have the time.

This time is different. At this point in my life—I'm fifty-five as I write this—I think it's time to set the record straight. So much of the information out there about me is wrong. I've come to hate books about Apple and its history so much because of that. For instance, there are stories that I dropped out of college (I didn't) or that I was thrown out of the University of Colorado (I wasn't), that Steve and I were high school classmates (we were several years apart in school) and that Steve and I engineered those first computers together (I did them alone).

Of course I understand that inaccuracies and rumors happen when you're in the public eye. And I even have a good insider's perspective as to how they happen. A perfect example of this, which I mentioned before, is when I was leaving Apple to start CL 9, my remote control company, in the late 1980s. The *Wall Street Journal* reporter called me and asked me directly if I was leaving because I was unhappy at Apple, and I told him directly that, no, I wasn't leaving because I was unhappy at Apple. Though I mentioned there were a few problems with morale in my view, I explicitly told him the only reason I was leaving was that I wanted to start a new company. Not because of any problems. And in fact, technically, I wasn't leaving at all.

To this day, I'm an Apple employee—I still have my Apple Employee ID card—and I receive a very low salary. And I continue to represent Apple at events and at speeches.

But the article the paper printed was wrong in two key places: It said I left Apple, and it said I left because I was disgruntled. Both untrue!

But you know what? Both errors went down as history. I mean, pick up almost any book on Apple's history and you'll probably read that wrong version of my story. Everything else major newspapers or early books got wrong about me went down as history, too.

So that's what's been bothering me—the fact that no one has gotten the story straight about how I built the first computers at Apple and how I designed them, and what happened afterward. So I hope this book sets the record straight, finally.

And there's another reason I've written this book, though I didn't realize it until I was well into it. I'd like to give advice, for what it's worth, to kids out there who are like I was. Kids who feel they're outside the norm. Kids who feel it in themselves to design things, invent things, engineer things. Change the way people do things.

I've learned a lot of lessons over the years, and not all of them involved how to handle ex-wives. Ha. In fact, none of them did.

No, my advice has to do with what you do when you find yourself sitting there with ideas in your head and a desire to build them. But you're young. You have no money. All you have is the stuff in your brain. And you think it's good stuff, those ideas you have in your brain. Those ideas are what drive you, they're all you think about.

But there's a big difference between just thinking about inventing something and doing it. So how do you do it? How do you actually set about changing the world?

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Well, first you need to believe in yourself. Don't waver. There will be people—and I'm talking about the vast majority of people, practically everybody you'll ever meet—who just think in black-and-white terms. Most people see things the way the media sees them or the way their friends see them, and they think if they're right, everyone else is wrong. So a new idea—a revolutionary new product or product feature—won't be understandable to most people because they see things so black and white. Maybe they don't get it because they can't imagine it, or maybe they don't get it because someone else has already told them what's useful or good, and what they heard doesn't include your idea.

Don't let these people bring you down. Remember that they're just taking the point of view that matches whatever the popular cultural view of the moment is. They only know what they're exposed to. It's a type of prejudice, actually, a type of prejudice that is absolutely against the spirit of invention.

But the world isn't black and white. It's gray scale. *As an inventor, you have to see things in gray scale.* You need to be open. You can't follow the crowd. Forget the crowd. And you need the kind of objectivity that makes you forget everything you've heard, clear the table, and do a factual study like a scientist would. You don't want to jump to conclusions, take a position too quickly, and then search for as much material as you can to support your side. Who wants to waste time supporting a bad idea? It's not worth it, that way of being stuck in your ego. You don't want to just come up with any excuse to support your way.

Engineers have an easier time than most people seeing and accepting the gray-scale nature of the world. That's because they already live in a gray-scale world, knowing what it is to have a hunch or a vision about what can be, even though it doesn't exist yet. Plus, they're able to calculate solutions that have partial values—in between all and none.

The only way to come up with something new—something world-changing—is to think outside of the constraints everyone else has. You have to think outside of the artificial limits everyone else has already set. You have to live in the gray-scale world, not the black-and-white one, if you're going to come up with something no one has thought of before.

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Most inventors and engineers I've met are like me—they're shy and they live in their heads. They're almost like artists. In fact, the very best of them *are* artists. *And artists work best alone*—best outside of corporate environments, best where they can control an invention's design without a lot of other people designing it for marketing or some other committee. I don't believe anything really revolutionary has ever been invented by committee. Because the committee would never agree on it!

Why do I say engineers are like artists? Engineers often strive to do things more perfectly than even they think is possible. Every tiny part or line of code has to have a reason, and the approach has to be direct, short, and fast. We build small software and hardware components and group them into larger ones. We know how to route electrons through resistors and transistors to make logic gates. We combine a few gates to make a register. We combine many registers to make an even larger one. We combine logic gates to make adders, and we combine adders to create others that can be used to create an entire computer. We write tiny bits of code to turn things on and off. We build upon and build upon and build upon, just like a painter would with colors on a paintbrush or a composer would with musical notes. And it's this reach for perfection—this striving to put everything together so perfectly, in a way no one has done before—that makes an engineer or anyone else a true artist.

Most people don't think of an engineer as an artist, probably, because people tend to associate engineers just with the things we create. But those things wouldn't work, they wouldn't be elegant or beautiful or anything else, without the engineer carefully thinking it out—thinking how to create the best possible end result with the fewest number of components. That's sophistication.

In my entire life I've only seen about one in twenty engineers who really exemplify that artistic perfection. So it's pretty rare to make your engineering an art, but that's how it should be.

I was very touched recently by a scene in the movie *Walk the Line*. In it, a producer tells Johnny Cash to play a song the way he would if that one song could save the whole world.

That line summed up a lot of what I look for when I talk about art in engineering or in anything.

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If you're that rare engineer who's an inventor and also an artist, I'm going to give you some advice that might be hard to take. That advice is: *Work alone.*

When you're working for a large, structured company, there's much less leeway to turn clever ideas into revolutionary new products or product features by yourself. Money is, unfortunately, a god in our society, and those who finance your efforts are businesspeople with lots of experience at organizing contracts that define who owns what and what you can do on your own.

But you probably have little business experience, know-how, or acumen, and it'll be hard to protect your work or deal with all that corporate nonsense. I mean, those who provide the funding and tools and environment are often perceived as taking the credit for inventions. If you're a young inventor who wants to change the world, a corporate environment is the wrong place for you.

You're going to be best able to design revolutionary products and features if you're working on your own. Not on a committee. Not on a team. That means you're probably going to have to do what I did. Do your projects as moonlighting, with limited money and limited resources. But man, it'll be worth it in the end. It'll be worth it if this is really, truly what you want to do—invent things. If you want to invent things that can change the world, and not just work at a corporation working on other people's inventions, you're going to have to work on your own projects.

When you're working as your own boss, making decisions about what you're going to build and how you're going to go about it, making trade-offs as to features and qualities, it becomes a part of you. Like a child you love and want to support. You have huge motivation to create the best possible inventions—and you care about them with a passion you could never feel about an invention someone else ordered you to come up with.

And if you don't enjoy working on stuff for yourself—with your own money and your own resources, after work if you have to—then you definitely shouldn't be doing it!

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It's so easy to doubt yourself, and it's especially easy to doubt yourself when what you're working on is at odds with everyone else in the world who thinks they know the right way to do things. Sometimes you can't prove whether you're right or wrong. Only time can tell that. But if you believe in your own power to objectively reason, that's a key to happiness. And a key to confidence. Another key I found to happiness was to realize that I didn't have to disagree with someone and let it get all intense. If you believe in your own power to reason, you can just relax. You don't have to feel the pressure to set out and convince anyone. So don't sweat it! You have to trust your own designs, your own intuition, and your own understanding of what your invention needs to be.

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If you could easily predict the future, inventing things would be a lot easier! Predicting the future is difficult even if you're involved with products that are guiding computers, the way we were at Apple.

When I was at Apple in the 1970s and 1980s, we would always try to look ahead and see where things were going. It was actually easy to see a year or two ahead, because we were the ones building the products and had all these contacts at other companies. But beyond that, it was tough to see. The only thing we could absolutely rely upon had to do with Moore's Law—the now-famous rule in electronics (named for Intel founder Gordon Moore) that says that every eighteen months you can pack twice the number of transistors on a chip.

That meant computers could keep getting smaller and cheaper. We saw that. But we had a hard time imagining what kinds of applications could take advantage of all this power. We didn't expect high-speed modems. We didn't expect computers to have large amounts of hard-disk storage built in. We didn't see the Internet growing out of the ARPANET and becoming accessible to everyone. Or digital cameras. We didn't see any of that. We really could only see what was right in front of us, a year or two out, max.

But there was one exception. Right around 1980, Steve and a bunch of us from Apple got to tour the Xerox Palo Alto Research Center (PARC) facility, which is one of Xerox's research and development labs.

Inside, for the first time ever, we saw real video displays—computer monitors—and they were showing something entirely new. They were showing the first graphical user interface (GUI)—an interface that lets you interact with icons and menus to control a program.

Up to this point, everything had been text-based. That's going to sound odd to all the people who don't remember it, but that's how everything worked back then. A computer user had to actually type in text commands—long, complicated ones—to make something happen.

But this experimental Xerox computer had windows popping up all over the place. And they were using this funny-looking device everyone now knows as a mouse, clicking on words and small pictures, the icons, to make things happen.

The minute I saw this interface, I knew it was the future. There wasn't a doubt in my mind. It was like a one-way door to the future—and once you went through it, you could never turn back. It was such a huge improvement in using computers. The GUI meant you could get a computer to do the same things it could normally do, but with much less physical and mental effort. It meant that nontechnical people could do some pretty powerful things with computers without having to sit there and learn how to type in long commands. Also, it let several different programs run in separate windows at the same time. That was powerful!

A few years later, Apple designed the Lisa computer, and later the Macintosh, around this concept. And Microsoft did it a couple years after that with Microsoft Windows. And now, more than twenty-five years after we saw that experimental computer in the Xerox PARC lab, all computers work like this.

It's so rare to be able to see the future like that. I can't promise it'll happen to you. But when you see it, you know it. If this ever happens to you, leap at the chance to get involved. Trust your instincts. It isn't often that the future lets you in like that.

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It's funny. In some ways, Apple is the bane of my life. That's because I'm hounded by people all the time—it's as if my whole life is constantly being directed by Apple's worldwide fame. But there was a time in the late and mid-1990s when it looked as if

Apple was in trouble. At least that's what all the media was reporting. That was shocking to me. Like most things in the world, the perception was pushed by the mass media and people's psychology. People would read things saying Apple was in trouble and the whole situation would just feed on itself. After reading stories like that, people were afraid to buy Apple products. I mean, at the time there were a lot of people going to Apple-using companies and schools and demanding that they switch over from their Macs to the new PCs. They were worried Macs weren't going to be around anymore. I was stunned by what was happening to Apple.

During the time Apple was supposedly going under, Gil Amelio was the CEO. He realized that the answer was to tighten up, start making more accurate amounts of product according to what we were going to sell, tighten our belt and restore profitability. But there was another problem. Macs running the operating system at the time, Mac OS 7, were crashing a lot. The sense that Macs running this operating system were weak and unreliable was a widely held belief throughout the Mac community: among the users, the executives, the employees—everyone. So one of the other things Apple decided was that it was going to need a new operating system.

At the time, this was an issue that meant a whole lot to me. I felt that Apple didn't need a new operating system. I felt that the current one was great—it was invulnerable to hackers and viruses, for one thing. I ran a major network in my home, and I never even once needed a firewall. I was as aware as everyone else of the Macs crashing, but I felt that fixing the current operating system would be a far better solution than finding a whole new one. And then one night, by accident, I figured out what the problem was. It was thanks to my son Jesse, who always likes to think different and not use the mainstream products that are out there. He downloaded a web browser called iCab and was using it instead of Internet Explorer (IE). Because he was using it, I gave it a try. And I fell in love with it! That first day I used iCab instead of IE, I had no crashes. Not a single one. Hmm. That night in bed, I lay there and wondered what the heck was up. And the next day my Mac didn't crash. I went two weeks before I had to restart the system, and that was a record!

From then on I realized that for the most part I had no crashes at all, and the only thing I'd changed about my system was I'd stopped using IE. I realized that by this time, almost everyone who had a Mac was running IE. That's why there were so many crashes, in my opinion. And it turned out that the reason neither Apple nor anyone else believed me was that the error in IE causing the crashes didn't just happen when you had IE open on your screen, it could happen anytime your computer was running. So it wasn't easy for them to see that it was IE, and not the system, causing the crashes.

As soon as I figured this out, I informed Apple at every possible level. I told every Apple employee and executive I knew. But no one would listen. What's funny is, at the time I did have a few friends who said their Macs never crashed. I figured they were either babying their computers and not really using them, turning them off every night, or lying. But now I asked them what browser they were using, and all my friends who claimed their Macs never crashed said they were using Netscape, which was another browser on the market at the time. I started going online to email lists and asking people what browser they were using. And yep, everyone who didn't have crashes was using Netscape.

I could never convince Apple. This was such a big lament for me at the time. I couldn't convince anyone that it wasn't the Mac OS that was at fault.

Then one day Gil Amelio told me that Apple—in addition to avoiding excess production and inventory and keeping expenses down—was going to buy a new operating system. They were going to buy the operating system from NeXT, which was the company Steve Jobs started after he left Apple.

Gil called me and said, "Steve, I'm letting you know that we're doing a deal with NeXT for $400 million." Wow, was I stunned! I never expected that. And I knew this meant Steve Jobs was coming back, which I'd also never expected! I knew that many people at Apple felt Steve had been disloyal by leaving Apple in 1985. (Steve resigned after a power struggle with the board. They stripped Steve of most of his responsibilities and Steve quit. It's a misconception in Silicon Valley that he was fired. He quit. And that made him look disloyal.)

It turned out that Steve, who just came back as an adviser at first, was exactly what Apple needed. I mean, a company like Apple largely depends on strong passions and the commitment of its customers. Apple became very passionate when its whole success and survival was questioned. The threat to its whole existence was so extreme! But Steve was able to stand up there on a stage and talk about Apple and really restore the loyalty that people had all along. Apple needed marketing leadership and charisma to get people excited again, and that's what Steve Jobs brought when he came back.

It's funny, because the products people credit with bringing Apple back to life—the iPods and the iMacs—all of them were in the design phase back when Apple was in trouble. Their main designer, Jonathan Ive, was already working on them. But the way Steve presented those new products was amazing. He made sure the press leaks were cut down, too, so when these new products came out—the colorful iMacs and, of course, the digital music iPods—they seemed to be totally new and surprising.

To be honest, I was never all that crazy for the iMacs. I had my doubts about its one-piece design. I didn't care about its colors and I didn't think its looks were all that good. It turned out that I just wasn't the right customer for it. Boy, it turned out to be the perfect product for schools—a low-cost, one-piece Macintosh.

And then there was the iPod. Now, you have to understand that, for me, portable music has always been important. Ever since my first transistor radio, I always had music I could carry with me. I was always the one to have the first portable tape players, the first portable CD players. I was the first person I knew with a minidisc player. And during my trips to Japan, where you always see products way ahead of their time and whatever is currently available here, I saw small players that could store music on memory chips. Essentially these devices stored music on little cards with RAM, the same kinds of cards your digital camera stores photos on. I would always buy whatever cool thing I could find there.

So when the iPod came out, I was excited. It was on the expensive side—it had a little disk drive built in—but it turned out that was the way to do it. Steve was always good at that. He really is great at looking at new technologies and choosing the right ones, the ones that will succeed.

The engineer in me wants to use and test a product before judging it. I compared my first iPod to my then-favorite, with no moving parts, the Diamond Rio 500. There's something pure about having no moving parts other than electrons. But the Diamond Rio took about a thousand dollars in memory cards to store the music I wanted for a plane flight. I also compared it to my Sony MiniDisc player, which I carried on every trip. With the MiniDisc, it was cheap to record your own music. Just putting these machines side by side, I couldn't say the iPod was superior. But after using the iPod on one plane flight, something about it felt natural and intuitive. It felt so right, and I knew I'd never go back to the other players. Now I can see that the iPod has changed the world. There's no doubt about it. I think it was the first special thing to happen to music since the Sony Walkman (portable music player) came out. And the iPod had something even bigger going for it. Unlike other MP3 players that were coming out at the time, it had its own software that Apple designed (iTunes) and it treated your computer as the center of things. The computer, your main computer, is where all the music can really be stored. The iPod is really a satellite. And that turned out to be the perfect way to do things.

It was exactly the right paradigm. And wow, it makes so much sense that Apple was the one to come up with it. After all, Apple's whole history is making both the hardware and the software, with the result that the two work better together. That's why Apple computers historically worked better than IBM-compatible PCs, where any company could make the hardware and someone else would make the software. So with the iPod, Apple made iTunes the software and iPod the hardware. They work together as one. Amazing! It's only because Apple supplied both sides of the equation—the hardware and the software—that it was able to create a product as great as this.

I'm proud now. I'm especially proud not just because Apple turned around, but because it turned around in a way so in line with our early values. Those values were about excellence in product design—so excellent that people would drool over the idea of having that product. Those values were about an emotional feeling—a feeling of fun. Like the way we decided to have color on the Apple II in the early days, back when nobody thought it would happen. I'm so proud that Apple has gotten back to the important things.

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If you're as lucky as I've been, then you'll get to live in a time when you're young just as a revolution is about to take off. Just like Henry Ford was there for the automotive industry, I was there to see and build the first personal computers.

Back in the mid-1990s when I was teaching school, I thought one time to myself, Wow, I wish I could be twelve now, look at the things I could do with what's out there now.

But then I realized I was lucky. I got to see the before, the during, and the after of some of those changes in life. I got to be one of those few people who could effect some of those changes.

Excellence came to me from not having much money, and also from having good building skills but not having done these products before.

I hope you'll be as lucky as I am. The world needs inventors—great ones. You can be one. If you love what you do and are willing to do what it takes, it's within your reach. And it'll be worth every minute you spend alone at night, thinking and thinking about what it is you want to design or build. It'll be worth it, I promise.